Practice Set A

Class 12 Mathematics, Chapter 12: Linear Programming

Maximum marks: **27** **10** questions

Name _____ Date _____ Score _____ / 27

1. All questions are compulsory. Marks are shown against each question.
2. Show every step of working: steps carry marks, not just the final answer.
3. Where a question offers OR, attempt only one of the two choices.

Q No.	Question	Marks

Level 1

4 marks

1 MCQ, CBSE SQP 2025-26 — 1 mark

If $Z = ax + by + c$, where $a, b, c > 0$, attains its maximum value at two of its corner points $(4, 0)$ and $(0, 3)$ of the feasible region determined by a system of linear inequalities, then:

(A) $4a = 3b$ **(B)** $3a = 4b$

(C) $4a + c = 3b$ **(D)** $3a + c = 4b$

2 MCQ, CBSE SQP 2025-26 — 1 mark

The graph of the inequality $3x + 5y < 10$ is the:

(A) entire XY-plane

(B) open half plane that does not contain the origin

(C) open half plane that contains the origin, but not the points of the line $3x + 5y = 10$

(D) half plane that contains the origin and the points of the line $3x + 5y = 10$

3 MCQ, CBSE 2025 — 1 mark

A factory produces two products X and Y. The profit earned by selling X and Y is represented by the objective function $Z = 5x + 7y$, where x and y are the number of units of X and Y respectively sold. Which of the following statements is correct?

(A) The objective function maximizes the difference of the profit earned from products X and Y.

(B) The objective function measures the total production of products X and Y.

(C) The objective function maximizes the combined profit earned from selling X and Y.

(D) The objective function ensures the company produces more of product X than product Y.

Q No.	Question	Marks
4	Assertion-Reason, CBSE 2025 **Assertion (A):** Every point of the feasible region of a Linear Programming Problem is an optimal solution. **Reason (R):** The optimal solution for a Linear Programming Problem exists only at one or more corner point(s) of the feasible region. **(A)** Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A). **(B)** Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A). **(C)** Assertion (A) is true, but Reason (R) is false. **(D)** Assertion (A) is false, but Reason (R) is true.	1 mark
Level 2		**9 marks**
5	Short answer, PYQ-style Solve the following linear programming problem graphically: Minimise $z = 6x + 3y$ subject to the constraints $4x + y \geq 80$, $x + 5y \geq 115$, $3x + 2y \leq 150$, $x \geq 0$, $y \geq 0$.	3 marks
6	Short answer, CBSE SQP 2024-25 Consider the following linear programming problem: Minimise $Z = x + 2y$ subject to $2x + y \geq 3$, $x + 2y \geq 6$, $x, y \geq 0$. Show graphically that the minimum of Z occurs at more than two points.	3 marks
7	Short answer, CBSE SQP 2025-26 Solve graphically: Maximise $Z = 2x + y$ subject to $x + y \leq 1200$, $x + y \geq 600$, $y \leq \frac{x}{2}$, $x \geq 0$, $y \geq 0$.	3 marks
Level 3		**14 marks**
8	Long answer, PYQ-style Solve the following linear programming problem graphically: Minimise and maximise $Z = 3x + 9y$ subject to the constraints $x + 3y \leq 60$, $x + y \geq 10$, $x \leq y$, $x \geq 0$, $y \geq 0$.	5 marks
9	Long answer, PYQ-style A manufacturing company makes two types of teaching aids A and B of Mathematics for Class XII. Each type A requires 9 labour hours for fabricating and 1 labour hour for finishing. Each type B requires 12 labour hours for fabricating and 3 labour hours for finishing. For fabricating and finishing, the maximum labour hours available per week are 180 and 30 respectively. The company makes a profit of ₹80 on each piece of type A and ₹120 on each piece of type B. Formulate this as an LPP and solve it graphically to find how many pieces of each type should be made per week to get the maximum profit. Also find the maximum profit.	5 marks

Q No.	Question	Marks
10	Case study, Board-pattern **Case Study:** A dairy farmer mixes two brands P and Q of cattle feed. One bag of brand P contains 3 units of nutrient A, 2.5 units of nutrient B and 2 units of nutrient C. One bag of brand Q contains 1.5 units of nutrient A, 11.25 units of nutrient B and 3 units of nutrient C. The mixture must contain at least 18 units of A, 45 units of B and 24 units of C. Brand P costs ₹250 per bag and brand Q costs ₹200 per bag. Let x bags of P and y bags of Q be mixed. **(i)** Write the objective function and the constraints of the LPP in simplest form. — 1 mark **(ii)** Find the corner points of the feasible region. — 1 mark **(iii)** Find the minimum cost of the mixture, and justify why the minimum exists even though the feasible region is unbounded. — 2 marks	4 marks

Answer key

Chapter 12: Linear Programming, Practice Set A. Full step-wise solutions are in the solutions sheet.

Level 1

1	(A) $4a = 3b$
2	(C) open half plane that contains the origin, but not the points of the line $3x + 5y = 10$
3	(C) The objective function maximizes the combined profit earned from selling X and Y.
4	(D) Assertion (A) is false, but Reason (R) is true.

Level 2

5	Minimum $z = 150$ at $x = 15$, $y = 20$.
6	Minimum $Z = 6$ at every point of the segment joining $(0, 3)$ and $(6, 0)$, i.e. at more than two points.
7	Maximum $Z = 2400$ at $x = 1200$, $y = 0$.

Level 3

8	Minimum $Z = 60$ at $(5, 5)$; maximum $Z = 180$ at every point of the segment joining $(15, 15)$ and $(0, 20)$.
9	12 pieces of type A and 6 pieces of type B per week; maximum profit ₹1680.
10	(i) Minimise $Z = 250x + 200y$ subject to $2x + y \geq 12$, $2x + 9y \geq 36$, $2x + 3y \geq 24$, $x, y \geq 0$ (ii) $(0, 12)$, $(3, 6)$, $(9, 2)$, $(18, 0)$ (iii) ₹1950, with 3 bags of P and 6 bags of Q